Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/31/2025
PI Name:	Haus, Miranda Ph.D.		
Project Title:	The Sources and Extent of Root Stunting During Growth in Lunar Highland Regolith and Its Impact on Legume Symbioses		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBR: E.9 Space Biology Research Studies
Start Date:	02/08/2024	**End Date:**	02/07/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/07/2026 per AM Ruby/NASA KSC (Ed., 3/21/25).		
Key Personnel Changes/Previous PI:	Michelle Thompson and Briony Horgan are collaborators.		
COI Name (Institution):	Wilhelm, Roland Ph.D. (Purdue University)		
Grant/Contract No.:	80NSSC24K0685		
Performance Goal No.:			
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Task Description:	Root growth is fundamental to plant productivity, regulating the growth and quality of the aboveground shoots. Roots respond dynamically to their rooting medium and these interactions can determine their capacity to acquire nutrients, resist toxicity, and, in soils on Earth, promote microbiological activity. Plants can grow in fertigated lunar regolith but their fitness is diminished by apparent root stressors that cause wound response, resulting in necrotic / apoptotic cells and stunted or abnormal growth. The extent to which root wounding impacts overall plant fitness in regolith is not well understood. Wounding can cause root electrolyte leakage and systemic transpiration losses that sap plant vigor and exacerbate other stress responses. Yet wounded root cells also serve as infection sites for microorganisms – a phenomenon that could be exploited by supplying mutualistic endophytes to promote root regeneration. We propose to comprehensively study (i) the causes of root wounding and the extent of damage incurred during growth in lunar highland regolith, and (ii) the interaction between wounding and soil-borne endophytes. Specifically, our experiments with lunar regolith will (Aim 1): characterize root wound response and the extent of impacts on plant fitness; (Aim 2): link mineralogy and local conditions in the regolith with wound response; and (Aim 3): correlate wounding with colonization by endophytic bacteria and assess any improvements in root regeneration or plant performance. We will evaluate plant and root fitness by growing plants in regolith in rhizotrons, measuring root growth rates, root system architecture, and wounding patterns over time. To assess plant vigor, developmental phenotypes will be complemented by collecting physiological data. Plant stress response will be assayed by using stomatal conductance, hormone activity, root exudation, gene expression, and reactive oxygen species production, along with the metal uptake by roots (inductively coupled plasma mass spectrometry, or "ICP-MS"). Wounding hotspots in regolith will be non-destructively characterized with Fourier Transform Infrared (FTIR) spectroscopy and Raman spectroscopy. Root colonization efficiency will be assessed with fluorescent microscopy and gene expression. Both root and bacterial gene expression will be profiled with metatranscriptomics (all complete genomes are available). Our findings will advance understanding of the causation and extent of root wound response occurring in lunar regolith. We will generate data linking plant fitness, root behavior, wounding types, and the stress response of roots with the chemical and physical conditions of the regolith substrate. As a complement, our experiments will assess potential biological mitigation strategies, testing the degree to which soil-borne endophytes can offset wounding and the effects of regolith stressors. This knowledge will inform research questions about how wounding type and root leakage impact plant fitness, and guide reasoning on the best approaches to condition regolith for agriculture, including understanding microbial community interactions in regolith and exploring metal toxicity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our findings will advance understanding of the causation and extent of root wound response occurring in lunar regolith. As a complement, our experiments will assess potential biological mitigation strategies, testing the degree to which soil-borne endophytes can offset wounding and the effects of regolith stressors. This knowledge will inform research questions about how wounding type and root leakage impact plant fitness, and guide reasoning on the best approaches to condition regolith for agriculture, including understanding microbial community interactions in regolith and exploring metal toxicity.
Task Progress:	This project investigates how lunar regolith simulant affects root development and plant-microbe interactions, focusing on legume species like Medicago sativa (alfalfa). While plants can be grown in nutrient-supplemented lunar regolith, past studies have shown substantial stunting and stress. A central hypothesis of our proposed work is that physical and chemical stressors in regolith—such as abrasive particles and trace metals—induce a wound response in roots, impacting overall plant performance and creating new microbial entry points. Our experiments aim to identify the causes of root stunting, characterize wound responses, and assess the role of beneficial microbes in mitigating these effects. During the first year of this project, all plant growth was performed using surface-sterilized seeds and aseptic technique. These methods enabled the consistent comparison of root traits and physiological responses across treatments. In addition, the regolith simulant was used without any prior washing or autoclaving to preserve the geological integrity of the simulant—a decision that will allow more accurate correlations between root stress and substrate properties in future analysis. We developed a robust and sterile growth system using custom 3D-printed mini-rhizoboxes for high-resolution root imaging. These mini-rhizoboxes were used to grow M. sativa seedlings in two types of regolith simulant (JSC-1A and OPRH4W30), under controlled environmental conditions. Plants were imaged every four hours using flatbed scanners, generating a time-course dataset of root system development across treatments. These images are being analyzed using machine learning segmentation tools (RootPainter), followed by quantification of root architecture traits with Rhizovision and SmartRoot. As planned, we tested whether root wounding promotes colonization by endophytic microbes. We inoculated select rhizoboxes with either the nodulating bacterium Ensifer meliloti or the non-nodulating Paraburkholderia phytofirmans. These microbes are expected to interact with root tissues differently depending on root integrity, which may help us assess the severity and type of wounding. Samples will be evaluated using gene expression and microscopy in future phases of the project. To characterize physical stress and wound response, we performed glucose exudation assays using Ampliflu Red, allowing us to localize regions of the root with elevated release—indicative of wound leakage or oxidative stress. These regions were further imaged using confocal microscopy following tissue clearing with the ClearSee protocol. Samples were also collected for transcriptomic profiling. Preliminary sample preparation and freezing protocols were established, and RNA extraction is underway using Qiagen miniprep kits. Although several experiments are still in progress, our imaging and phenotyping efforts have already revealed a variety of root growth patterns, including altered root tip histology, aberrant root angles, and reduced elongation in regolith simulant conditions. These observations are consistent with prior findings on regolith-induced stress but extend them by showing spatial patterns in root development and wound site localization. By combining root architecture data with localized glucose release and confocal imaging, we are creating a multidimensional map of stress expression in the root system. These early results provide critical groundwork for linking specific regolith properties to plant responses. In the coming year, we will integrate root phenotypes with regolith simulant surface analysis (already underway using Visible and

	Near-Infrared/VNIR spectroscopy, scanning electron microscopy/SEM, and transmission electron microscopy/TEM) to determine if and how roots alter the chemical microenvironment around them. This will allow us to assess whether roots and microbes can “condition” regolith over time and improve its suitability for future crop growth. Our research will ultimately help NASA evaluate whether lunar regolith can support cyclical plant production, whether microbial partners can offset regolith-induced wounding, and how substrate properties and biological activity evolve together. These findings are foundational for developing regenerative agriculture systems for the Moon and other extraterrestrial environments. All root images, protocols, and sequencing data from this study will be deposited to the NASA GeneLab portal for public access and community use.
Bibliography Type:	Description: (Last Updated: 07/31/2025)
Conference Materials (Downloadable)	Miller G, Horgan B, Thompson M, Wilhelm R, Haus MJ. "Characterization of root wound response and its link with lunar regolith simulant properties." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024 2024_ASGSR_Poster (1).pdf